ANNUAL EXAMINATION - 2026-27

Blueprint

Chapter	Teaching Hours	Marks Weightage	Part A 1 mark 20 Qs (20 Marks)	Part B 2 marks 7 Qs (14 Marks)	Part C 3 marks 7Qs (21 Marks)	Part D 5 marks 10 Qs (50 Marks)	Total Questions
1. Living World	3	2	2	*	*	*	2
2. Biological Classification	5	4	1	*	1	*	2
3. Plant Kingdom	6	6	1	1	1	*	3
4. Animal Kingdom	9	8	1	1	*	1	3
5. Morphology of Flowering Plants	6	5	2	*	1	*	3
6. Anatomy of Flowering Plants	4	4	2	1	*	*	3
7. Structural Organisation in Animals	3	3	*	*	1	*	1
8. Cell: The Unit of Life	11	10	3	1	*	1	5
9. Biomolecules	7	6	1	*	*	1	2
10. Cell Cycle and Cell Division	6	5	*	1	1	*	2
11. Photosynthesis in Higher Plants	7	6	1	*	*	1	2
12. Respiration in Plants	7	6	1	*	*	1	2
13. Plant Growth and Development	8	7	*	1	*	1	2
14. Breathing and Exchange of Gases	6	5		1	1	*	2
15. Body Fluids and Circulation	7	6	1	*	*	1	2
16. Excretory Products & their Elimination	6	6	1	*	*	1	2
17. Locomotion and Movement	7	6	1	*	*	1	2
18. Neural Control and Coordination	5	4	1	*	1	*	2
19. Chemical Control & Coordination	7	6	1	*	*	1	2
Total	**120**	**105**	**20**	**5/7**	**5/7**	**5/10**	**35/44**

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MODEL PAPER

IPUC

Biology (36)

General Instructions:

1. The question paper consists of Parts, a, B, C, D and E.
2. Part-A consists of I and II and Part-D consists of V and VI.
3. All the parts are compulsory.
4. For Part-A questions, only the answers written first will be considered for evaluation.
5. Draw diagrams wherever necessary. Unlabelled diagrams do not attract any marks.

PART – A

I. Select the correct alternative from the choices given below: **$15 \times 1 = 15$**

1. Taxonomic hierarchy refer to

   a) Classification of species based on the fossil record

   b) A list of botanists or zoologists who have worked on the taxonomy of a species

   c) A group of senior taxonomists who decide the nomenclature of animals and plants

   d) Stepwise arrangement of all categories for classification of animals and plants

2. Choose the incorrect statement

   a) Methanogens are archaebacteria that produce methane in marshy areas.

   b) *Nostoc* is a filamentous blue green alga which fixes atmospheric nitrogen.

   c) Chemosynthetic autotrophic bacteria play a great role in recycling nutrients like nitrogen, phosphorous, iron and sulphur.

   d) Mycoplasma lacks a cell wall and can survive with oxygen

3. What distinguishes liverworts from mosses in particular?

   a) Mosses have vascular tissue that liverworts lack.

   b) In contrast to liverworts, mosses have an intermediate stage known as protonema.

   c) Mosses grow taller, whilst liverworts do not.

   d) Sporophytes, specialized bodies found only in mosses, are missing in liverworts.

4. An animal which can be included in phylum Arthropoda is

   a) *Aplysia* b) *Limulus* c) *Echinus* d) *Nereis*

5. Based on position of floral parts on the thalamus the figure depicts



a) Hypogynous b) Perigynous c) Epigynous d) Epiphyllous

6. **Statement I:** Endarch and exarch are the terms often used for describing the position of secondary xylem in the plant body.

   Statement II: Exarch condition is the most common feature of the root system.

   In the light of the above statements, choose the correct answer from the options given below:

   a) Both Statement I and Statement II are correct.

   b) Both Statement I and Statement II are incorrect.

   c) Statement I is correct but Statement II is incorrect.

   d) Statement I is incorrect but Statement II is correct.

7. Casparian strip is made up of

   a) Chitin b) Cellulose c) Pectin d) Suberin

8. On death of an animal cell it starts to degrade while the plant cell do not degrade the cell organelle involved is

   a) Vacuoles b) Chloroplast c) Lysosomes d) Ribosomes

9. Middle lamella of cell wall is mainly composed of

   a) Cellulose b) Calcium pectate c) Magnesium silicate d) Lignin

10. Match the proteins in column I with their functions in column II

Column I (Proteins)	Column II (Functions)
1. Trypsin	i. Intercellular ground substances
2. Insulin	ii. Enables glucose transport into cells
3. GLUT-4	iii. Hormone for conversion of glucose into glycogen
4. Collagen	iv. Proteolytic Enzyme

   a) 1-ii, 2-iv, 3-i, 4-iii b) 1-iv, 2-iii, 3-ii, 4-i

   c) 1-ii, 2-i, 3-iii, 4-iv d) 1-i, 2-iii, 3-ii, 4-iv

11. Read the following four statements, (i), (ii), (iii) and (iv) and select the right option having both correct statements.

   (i) Z scheme of light reaction takes place in presence of PSI only.

(ii) Only PSI is functional in cyclic photophosphorylation.

(iii) Cyclic photophosphorylation results into synthesis of ATP and NADPH2.

(iv) Stroma lamellae lack PSII as well as NADP.

a) (ii) and (iv) b) (i) and (ii) c) (ii) and (iii) d) (iii) and (iv)

12. In the subsequent pregnancy of a mother with foetus would cause severe anaemia and jaundice to the baby. This condition occurs when the blood group of mother, father & foetus are respectively

a) Rh+ve, Rh-ve & Rh+ve. b) Rh-ve, Rh-ve & Rh+ve

c) Rh+ve, Rh+ve & Rh+ve d) Rh-ve, Rh+ve & Rh+ve

13. GFR in a healthy individual is approximately.

a) 100ml/minute b) 125 ml/minute c) 200 ml/minute d) 500 ml/day

14. Choose the wrongly matched pair.

a) Isotropic band - Actin

b) Anisotropic band - Myosin

c) Central part of I- band - M-line

d) Central part of A-band - H-zone

15. A canal called the _____ passes through the mid brain.

a) Cerebral aqueduct b) Corpora quadrigemina c) Corpus callosum d) Cranial meninges

II. Fill in the blanks by choosing the appropriate word from those given in the bracket:

5 × 1 = 5

(Satellite, Isthmus, Aleurone, *Systema Naturae,* Pyruvic acid, Lactic acid)

16. Linnaeus used _____ as the title of his publication.

17. The proteinaceous part of the monocot seed is_____ .

18. The non staining secondary constriction on a chromosome is called _____.

19. During exercise when oxygen is inadequate for cellular respiration in muscle cells, the pyruvic acid is reduced to _____.

20. The two lobes of the thyroid gland are interconnected with a thin flap of connective tissue called ____

PART – B

III. Answer any five of the following questions in 2 to 3 sentences each, wherever applicable:

5 × 2 = 10

21. How are leaves in gymnosperms adapted to withstand extremes of climatic factors?

22. Give any two parasitic adaptations of flatworms.
23. Diagrammatically represent open and closed vascular bundles.
24. What are mesosomes? Mention any one function.
25. Mitosis is called as equational division. Justify
26. Explain the role abscisic acid as a a) Growth inhibitor b) Stress hormone.
27. Differentiate between vital capacity and total lung capacity.

PART – C

IVAnswer any five of the following questions in about 40-80 words each, wherever applicable:

5 × 3 = 15

28. Identify the class of fungi based on the examples given.
    i. *Alternaria*, *Colletotrichum*, *Trichoderma*
    ii. *Agaricus*, *Ustilago*, *Puccinia*
    iii. *Aspergillus*, *Claviceps*, *Neurospora*
29. Draw a neat labelled diagram showing different parts of a fern plant *Equisetum.*
30. Explain three different types of phyllotaxy in plants.
31. Explain the functions occurring in duodenum of the frog?
32. Write a note on Diplotene of Prophase I of Meiosis I.
33. Describe the role of haemoglobin in oxygen transport.
34. Explain the process of electrical synapse during transmission of nerve impulse.

PART – D

V. Answer any four of the following questions in 200–250 words each, wherever applicable.

4 × 5 = 20

35. Distinguish plant cell from animal cell.
36. Classify the various classes of enzymes based on type of reaction they catalyze.
37. Give schematic representation of Tricarboxylic acid cycle/Citric acid cycle.
38. Define growth rates. Give the mathematical expressions for linear and exponential growth and define the terms.
39. Write a note on neural & hormonal regulation of cardiac activity.
40. Draw a labelled diagram of L.S of human kidney.
41. Enlist the synovial joints and write the location of each joint.

VI. Answer any one of the following questions in 200–250 words each, wherever applicable:

$1 \times 5 = 5$

42. Each group of animals is different in many respects. at the same time, with in the groups also animals exhibit differences in their characters. in view of this, mention the special character for each of the following.

    a) *Torpedo* b) *Pleurobrachia* c) *Crocodilus* d) *Ichthyophis* e) *Ornithorhynchus*

43. Breakdown of proton gradient across the thylakoid membrane leads to synthesis of ATP. Describe how it occurs?

44. A sample of urine was diagnosed to contain a high content of glucose and ketone bodies. Based on this observation answer the following questions.

    a) Which endocrine gland and hormone is related to this condition? 2

    b) Name the cells on which this hormone acts. 1

    c) What is the condition called? 1

    d) How can it be treated? 1

PART – E

(For Visually Challenged)

23. Differentiate between open and closed vascular bundles.

29. How does sexual reproduction take place in pteridophytes?

40. Categorize the following animals based on the nitrogenous excretory products:

    a) Aquatic Amphibians b) Terrestrial Amphibians c) Reptiles

    d) Bony fishes e) Marine fishes

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